

Computational Systems Biology for Digital Medicine

7–12 December 2025

Wellcome Genome Campus, UK The Importance of Interoperability

Objectives

Objective 1: Search for a pathway of interest, download the file in the proper format (sbml, xml), and use the appropriate tool to open it, modify it, and save it (CellDesigner).

Objective 2: Use CaSQ to create an executable file (Boolean model) in different formats (sbml qual and JSON) and complementary files containing different information (sif files).

Objective 3: Import the sbml qual file in two different simulation platforms and perform preliminary experiments (simulations).

Objective 4: Import the JSON file to a different platform (BMA)

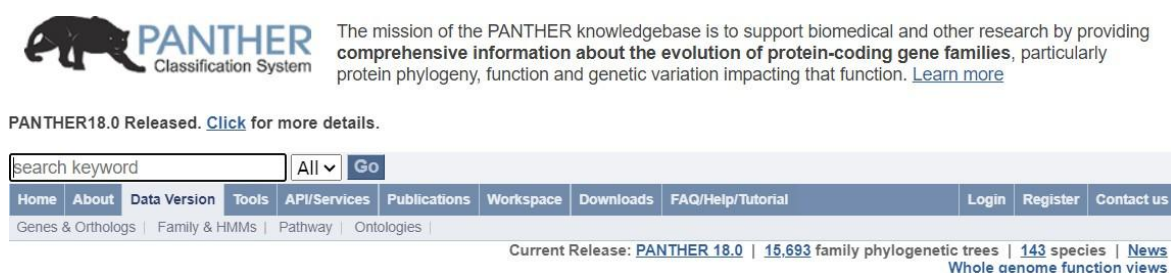
This tutorial links the different model types (static, process description, activity flow, dynamic, and executable) and the tools shown in the course. Each simulation platform has its own detailed tutorial. The purpose here is to focus on interoperability and provide participants with a quick guide to help them in their projects.

Finding a pathway of interest

Please go to the Pather DB database

<https://www.pantherdb.org/>

Then, search for the pathway list by clicking on the Data version tab



The mission of the PANTHER knowledgebase is to support biomedical and other research by providing comprehensive information about the evolution of protein-coding gene families, particularly protein phylogeny, function and genetic variation impacting that function. [Learn more](#)

PANTHER18.0 Released. [Click](#) for more details.

search keyword All

Home About Data Version Tools API/Services Publications Workspace Downloads FAQ/Help/Tutorial Login Register Contact us

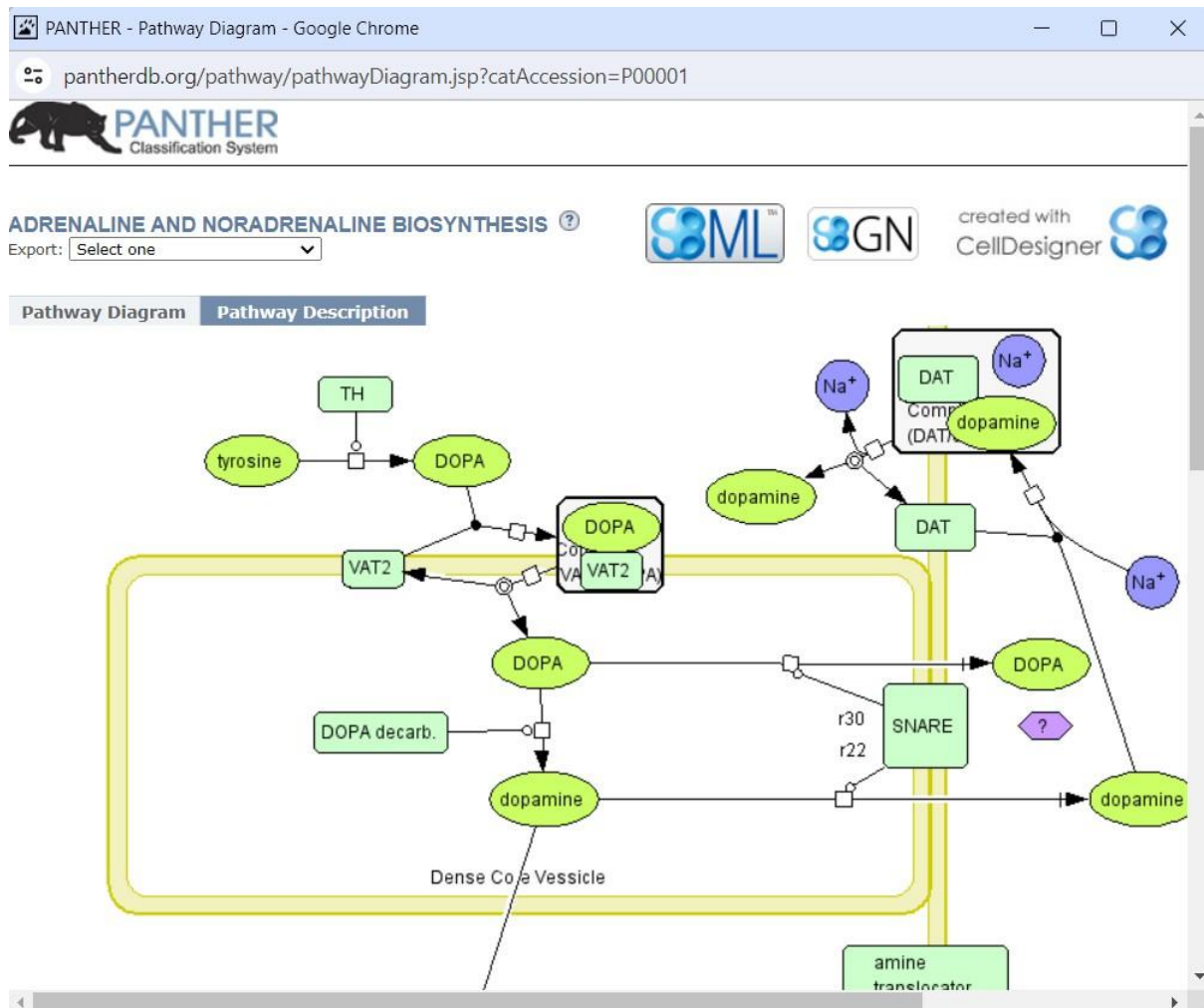
Genes & Orthologs | Family & HMMs | Pathway | Ontologies |

Current Release: [PANTHER 18.0](#) | [15,693](#) family phylogenetic trees | [143](#) species | [News](#)
[Whole genome function views](#)

Then click on Pathway

And subsequently, click on List of Pathways

Select the pathway Adrenaline and noradrenaline biosynthesis by clicking on it



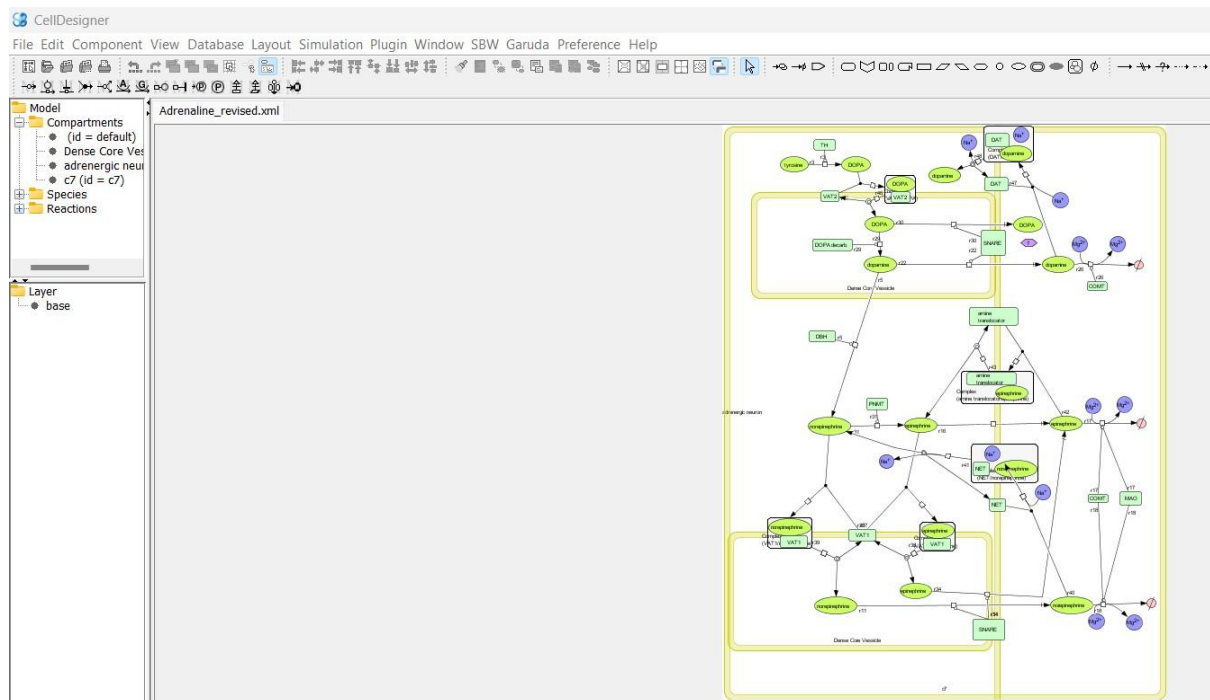
Spend some time to understand the pathway and read the pathway description

Export the pathway in SBML format and save it to your session.
You will notice that it has an xml suffix.

Visualising and modifying the diagram using Cell Designer

Open the Cell Designer tool and open the file.

Select a compartment and surround the graph. Save the file as Adrenaline_revised.xml to your session.



Creating an executable model

Now open your terminal and type

```
casq -s Adrenaline_revised.xml Adrenaline_revised.sbml
```

```
manager@CompSysBio25:~/Desktop$ casq -s Adrenaline_revised.xml Adrenaline_revised.sbml
manager@CompSysBio25:~/Desktop$
```

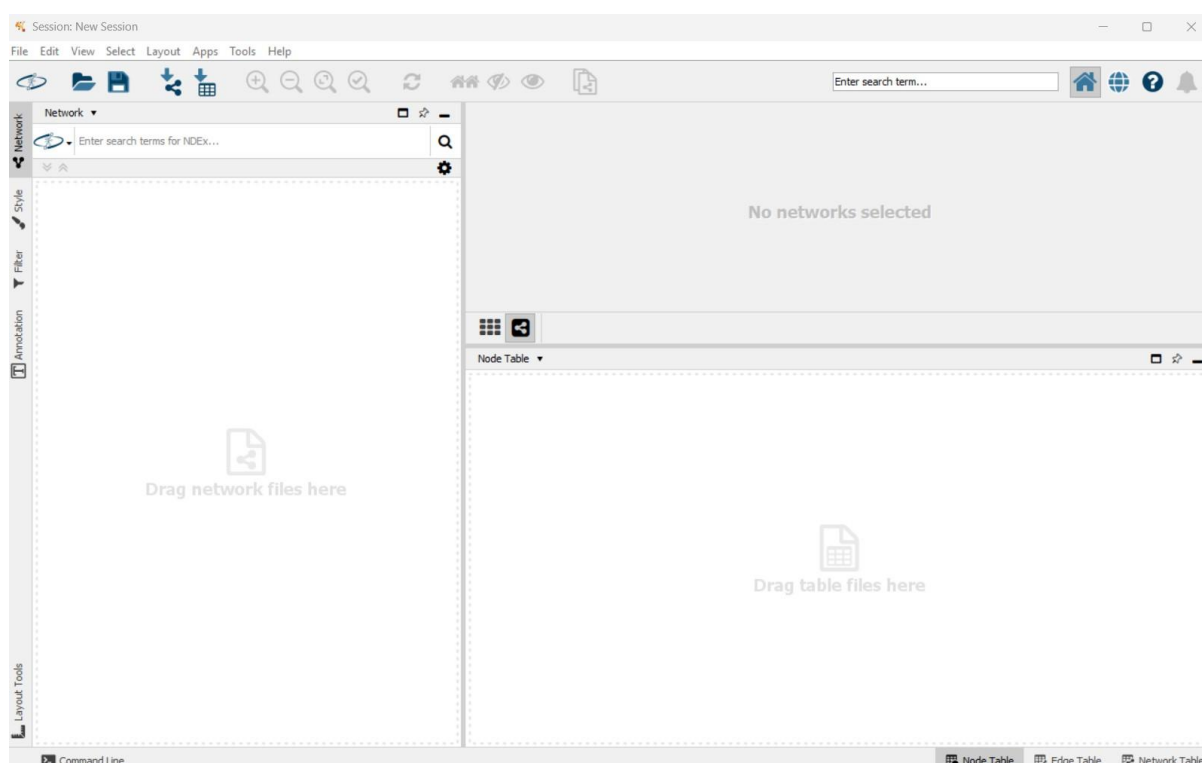
This will produce a series of files (sif, raw_sif, sbml).

Look at the sif files and determine what they contain (open them with a text editor).

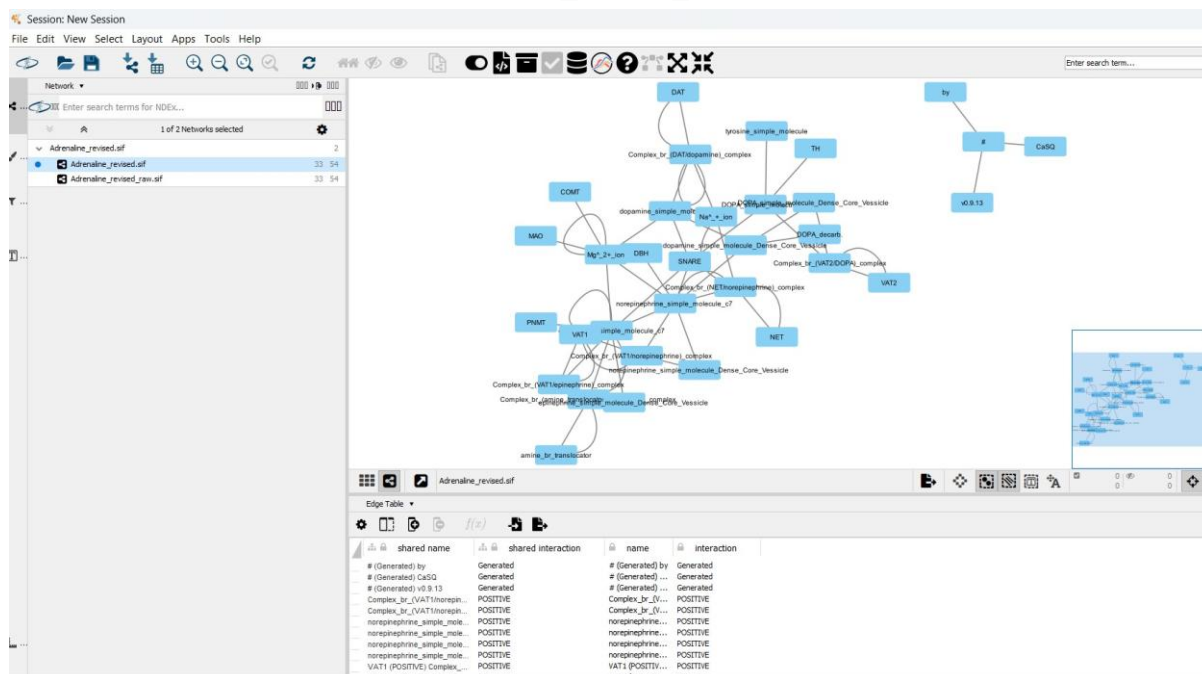
You can also import your sif files to Cytoscape. Open Cytoscape

(Open the terminal in your VM session, type Cytoscape and hit enter.

Cytoscape will launch a session.



Click Import network from file and select the Adrenaline_revised.sif file.
Then click again and import the Adrenaline_revised_raw.sif file.



What do you observe?

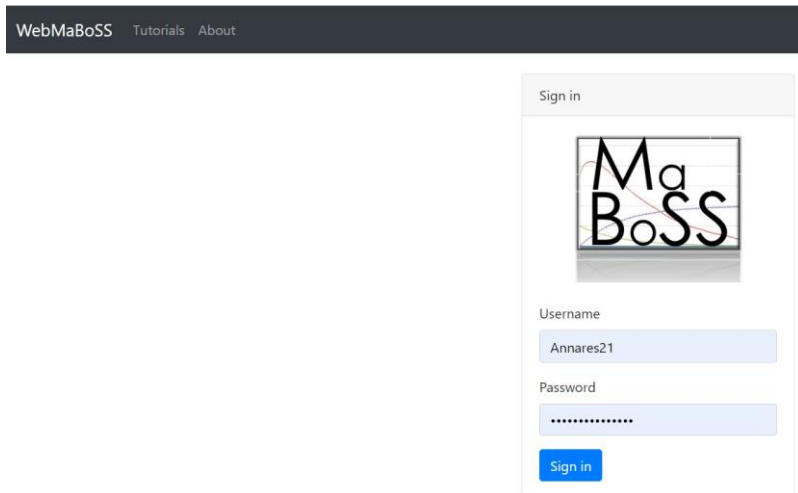
Using web-based simulation platforms to do *in-silico* experiments

WebMaBoSS

Go to the WebMaBoSS site:

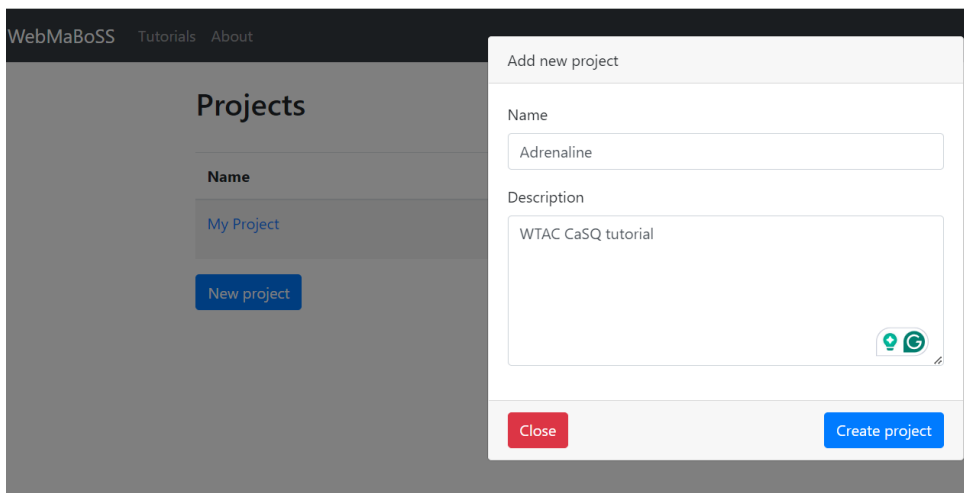
<https://webmaboss.vincent-noel.fr/login/>

Login to your session:



The login interface features a dark header with 'WebMaBoSS', 'Tutorials', and 'About' links. Below is a 'Sign in' modal window. It contains the WebMaBoSS logo, a 'Username' field with the text 'Annares21', a 'Password' field with masked characters, and a blue 'Sign in' button.

Create a project:



The 'Add new project' modal is overlaid on the 'Projects' page. The background page shows a 'New project' button. The modal contains a 'Name' field with 'Adrenaline', a 'Description' text area with 'WTAC CaSQ tutorial', and a small icon at the bottom right. At the bottom of the modal are 'Close' and 'Create project' buttons.

Now go to your newly created project space and load the Adrenaline_revised.sbml file

Models

Nothing to show

[Load model](#)
[Import model](#)

Click on Load model and then select SBML qual. Using the browse button, find your model and upload it—don't forget to name it!

Load model

Name

Type

SBML qual ▼

SBML file

Select file... Browse

☒ Use SBML names

Close

Load model

Load model

Name

Type

SBML qual ▼

SBML file

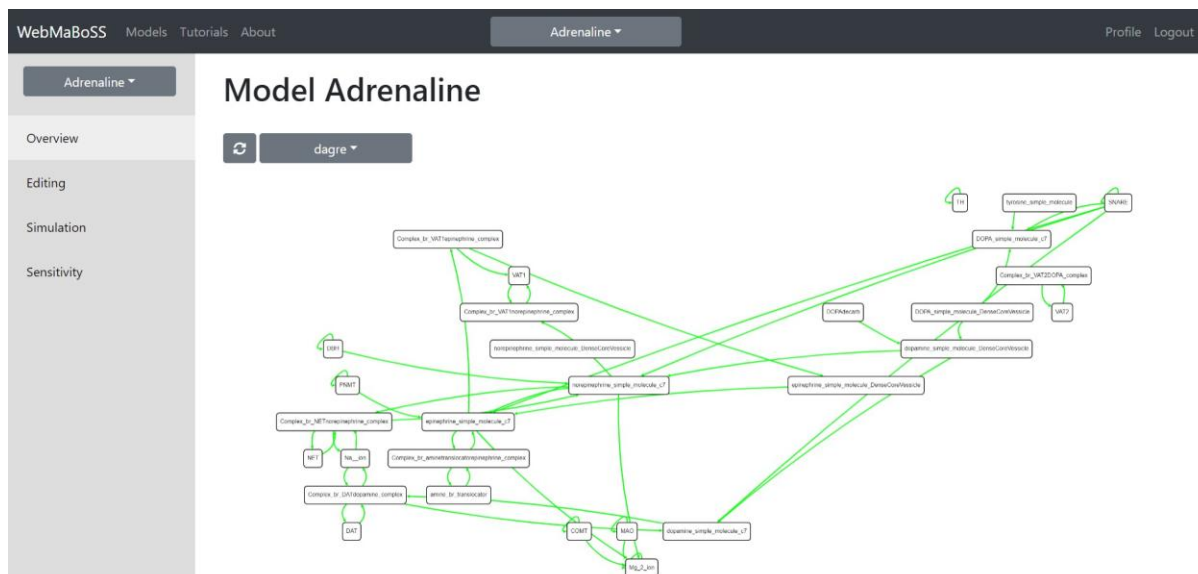
Adrenaline_revised.sbml Browse

☒ Use SBML names

Close

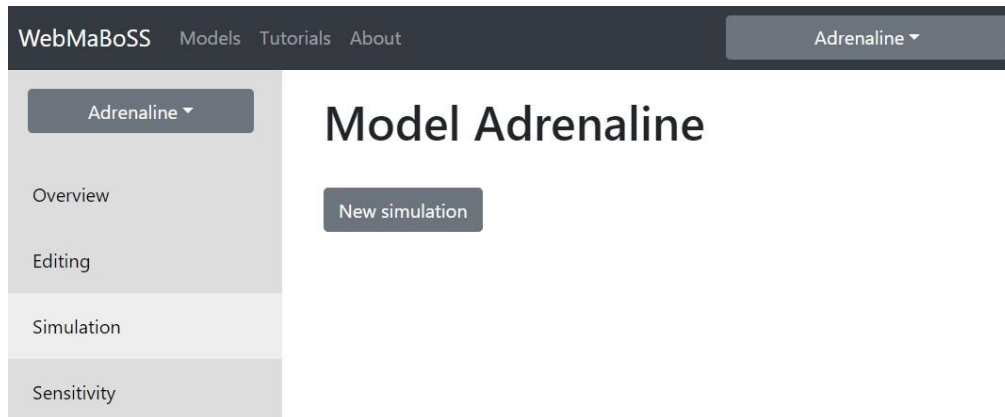
Load model

Now click on your model and see it



What do you observe? Compare this to the graph you see in Cell Designer (xml).
Compare this graph to the one you see in Cytoscape (sif).

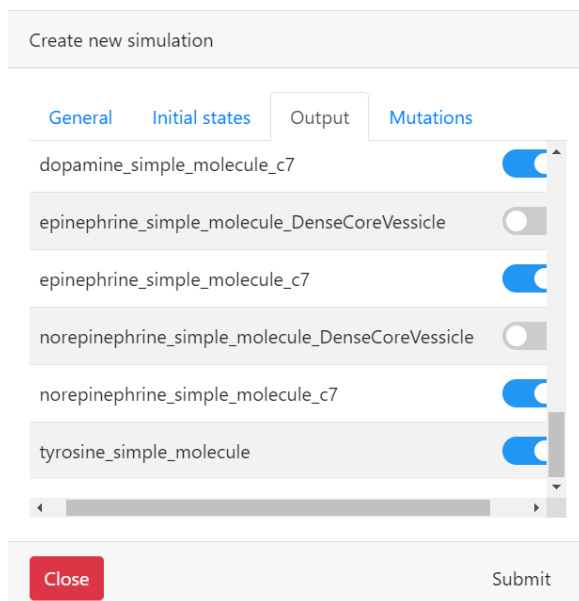
Now click on the simulation button and then to New simulation



The screenshot shows the WebMaBoSS interface for the 'Model Adrenaline'. The top navigation bar includes 'WebMaBoSS', 'Models', 'Tutorials', and 'About'. A dropdown menu on the right shows 'Adrenaline'. The left sidebar contains a menu with 'Overview', 'Editing', 'Simulation', and 'Sensitivity'. The main content area displays the title 'Model Adrenaline' and a 'New simulation' button.

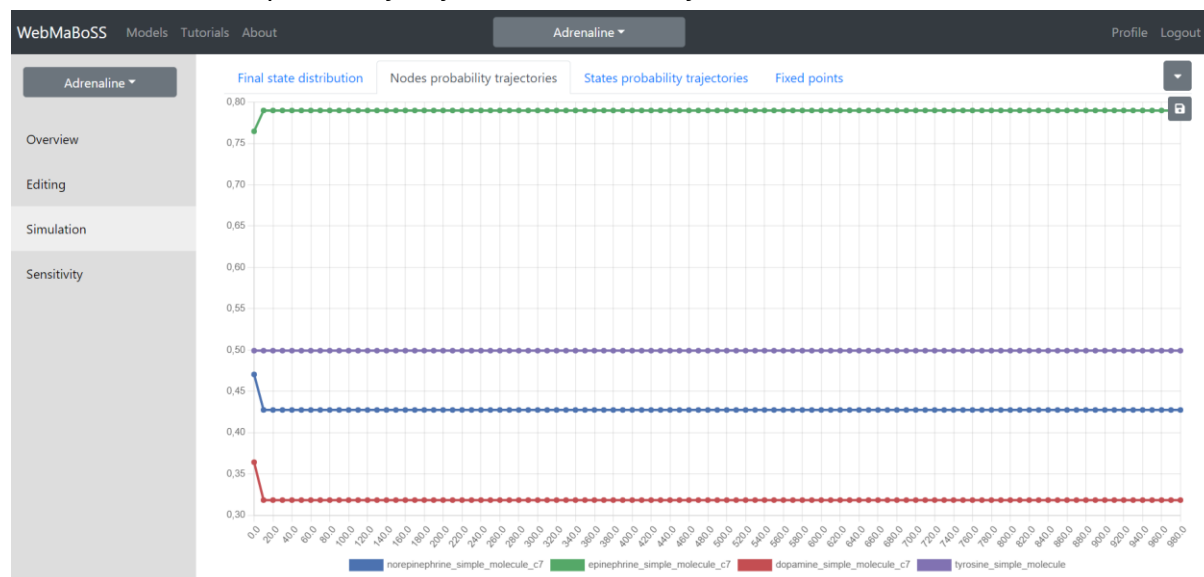
Leave the initial states at default configuration where all nodes have 50% activity and select as outputs only the four simple molecules:

Dopamine, norepinephrine, tyrosine, epinephrine. Then hit submit



The screenshot shows the 'Create new simulation' dialog box. It has four tabs: 'General', 'Initial states', 'Output', and 'Mutations'. The 'Output' tab is selected, showing a list of molecules with toggle switches. The molecules and their states are: dopamine_simple_molecule_c7 (on), epinephrine_simple_molecule_DenseCoreVessicle (off), epinephrine_simple_molecule_c7 (on), norepinephrine_simple_molecule_DenseCoreVessicle (off), norepinephrine_simple_molecule_c7 (on), and tyrosine_simple_molecule (on). At the bottom, there are 'Close' and 'Submit' buttons.

Observe the nodes' probability trajectories - what do you make of it?

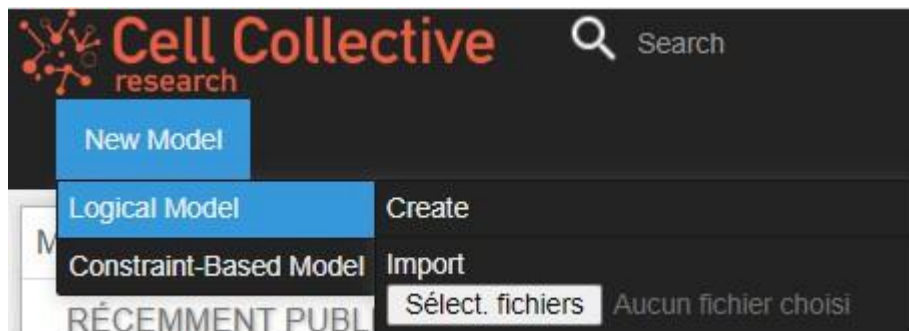


Cell Collective

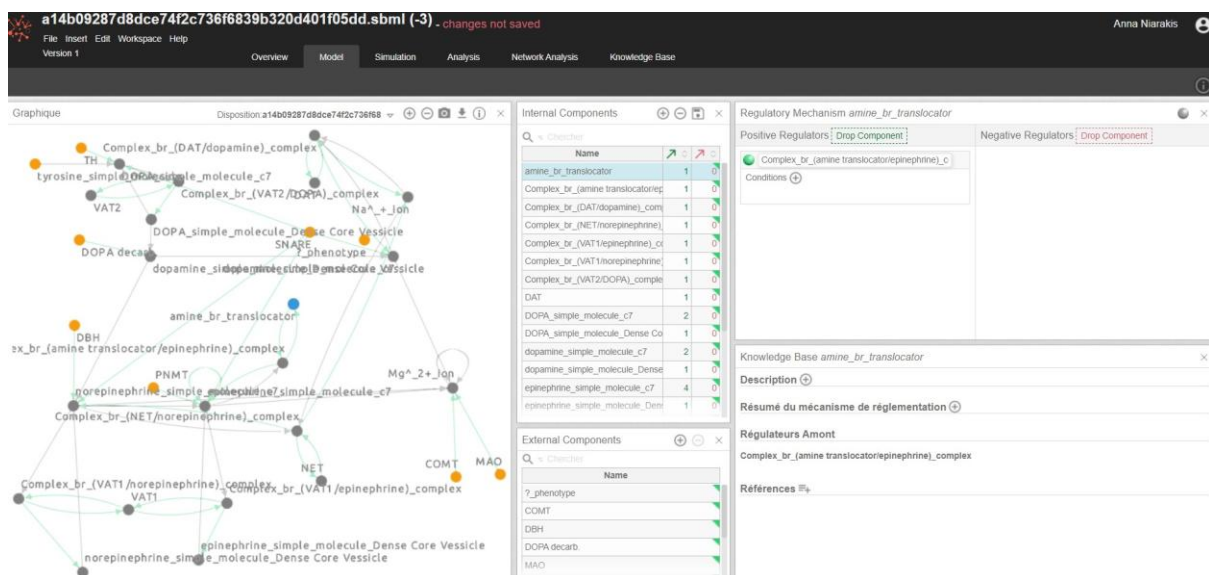
Login to your Cell Collective account

<https://research.cellcollective.org/?dashboard=true#>

Then click on the left upper panel New model, select Logical model and import

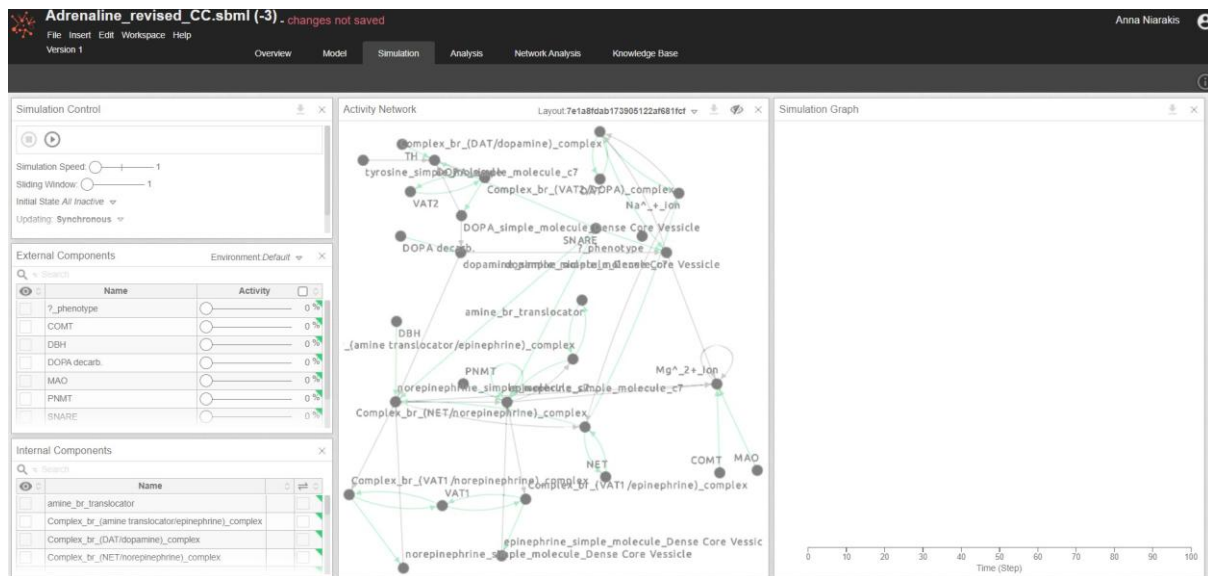


Choose your Adrenaline_revised.sbml file to import



Change the model name to Adrenaline_revised_CC and go to the Simulation tab





Here, you have a different way of simulating your model (detailed CC tutorial follows)
 In the simulation control panel, set the external components to 50% and click below the eye icon on the internal components for the simple molecules:
 Dopamine, norepinephrine, epinephrine, and under the external components for tyrosine

Simulation Control

Simulation Speed: 1
Sliding Window: 1
Initial State *All Inactive* ▾
Updating: *Synchronous* ▾

External Components

Environment: *New Env 1* ▾

Search

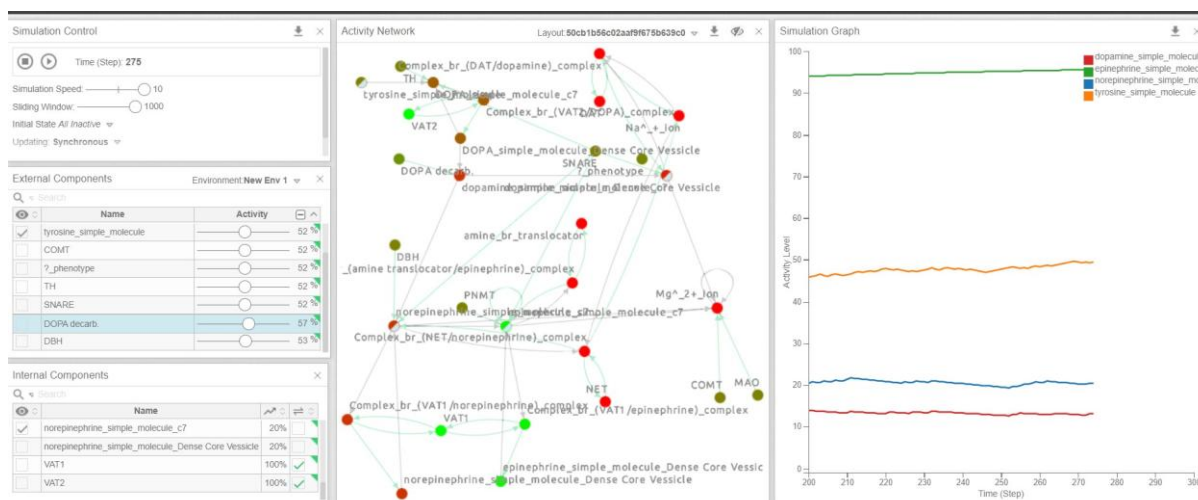
☐	Name	Activity	
☐	?_phenotype	<input type="range" value="50"/>	50 %
☐	COMT	<input type="range" value="50"/>	50 %
☐	DBH	<input type="range" value="50"/>	50 %
☐	DOPA decarb.	<input type="range" value="50"/>	50 %
☐	MAO	<input type="range" value="50"/>	50 %
☐	PNMT	<input type="range" value="50"/>	50 %
☐	SNARE	<input type="range" value="50"/>	50 %

Internal Components

Search

☐	Name		
☐	Na ⁺ _+_ion		☐
☐	NET		☐
☒	norepinephrine_simple_molecule_c7		☐
☐	norepinephrine_simple_molecule_Dense Core Vessicle		☐

Also, put as always active VAT1 and 2 and set the simulation speed and sliding window at max. Then launch the simulation.



What do you observe?

Are these results comparable with the ones obtained previously? Feel free to play around with the model and the platform.

BMA

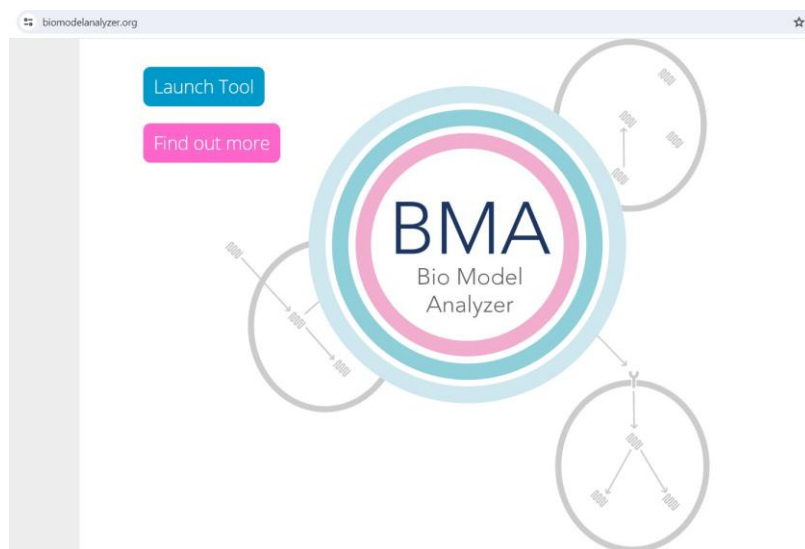
Now go back to your terminal and type
`casq -b Adrenaline_revised.xml Adrenaline_revised.json`

```
manager@CompSysBio25:~/Desktop$ casq -b Adrenaline_revised.xml Adrenaline_revised.json
manager@CompSysBio25:~/Desktop$
```

You should have now created your json file

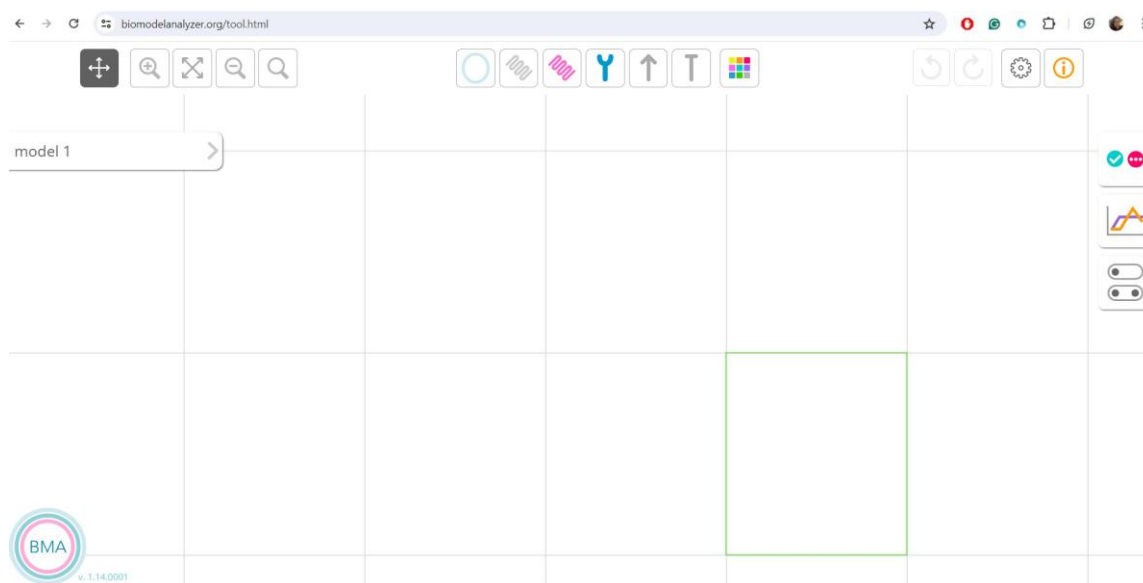
Go now to

<https://biomodelanalyzer.org/>

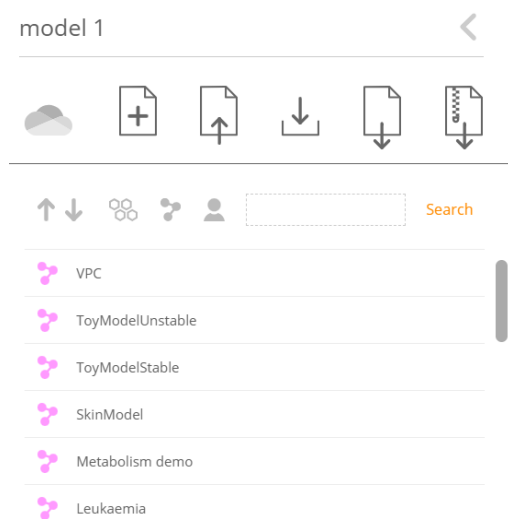


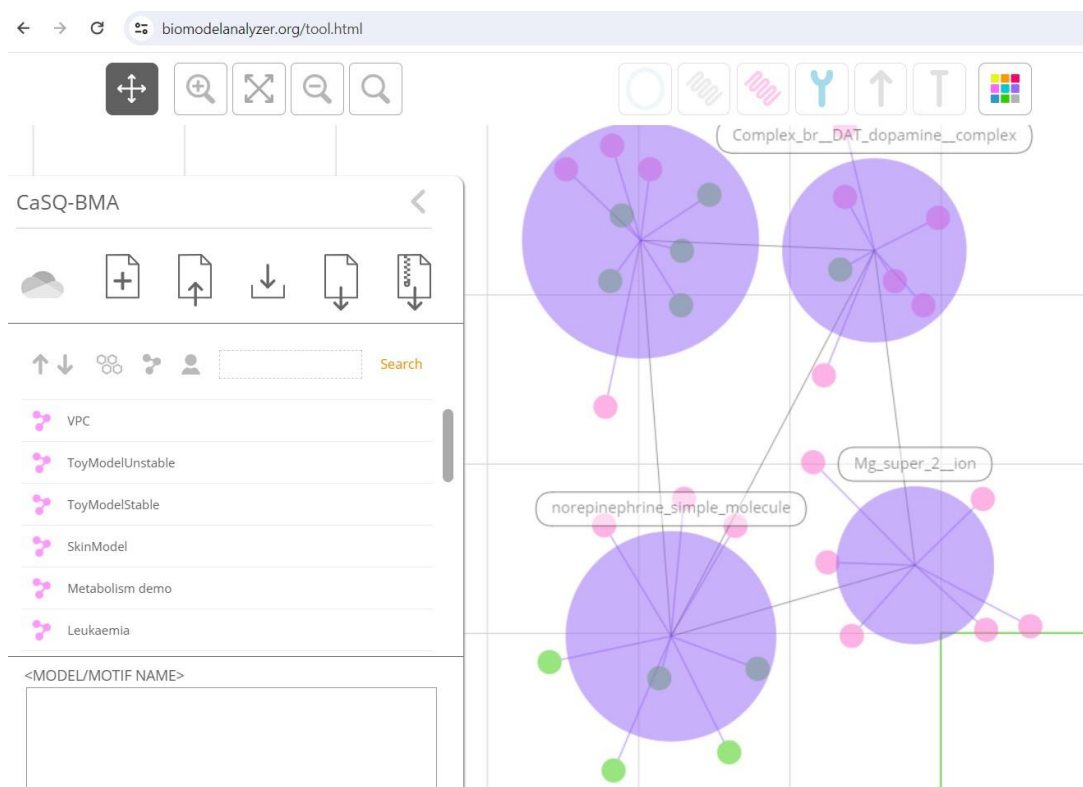
and launch the tool

Click on model1



And import your newly generated json file

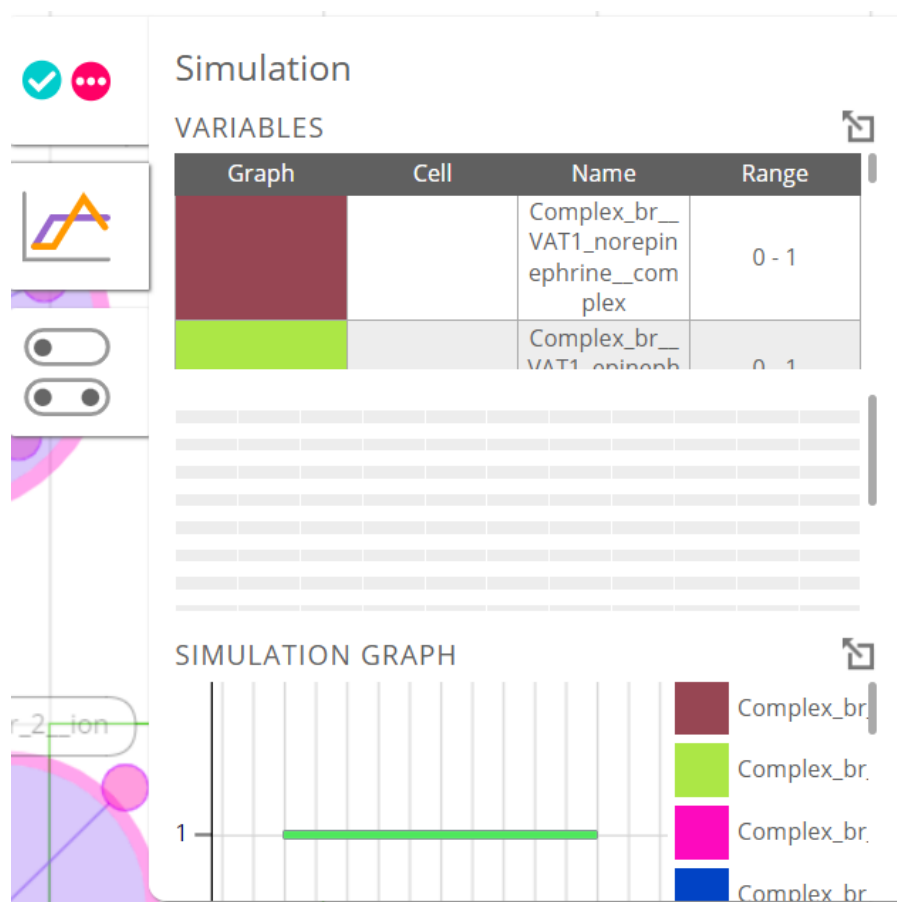




Click now on the simulation panel (the second button below)



The below panel will now appear on the screen:



Click on the arrow in the upper right on the variables panel

You can now imitate or randomize the initial conditions of the previous simulations.

Choose to see only the simple molecules in the graph

Here is an example:

Simulation Progression



 Randomise

Graph	Name	Range
	Complex_br_VAT1_...	0 1
	Complex_br_VAT1_...	0 1
	Complex_br_NET_n...	0 1
	Complex_br_amine...	0 1
	Complex_br_VAT2_...	0 1
	Complex_br_DAT_d...	0 1
	TH	0 1
	DOPA_decarb.	0 1
	DBH	0 1
	PNMT	0 1
✓	tyrosine_simple_mol...	0 1
	DOPA_simple_molec...	0 1
✓	dopamine_simple_m...	0 1
✓	norepinephrine_sim...	0 1
✓	epinephrine_simple_...	0 1
	NET	0 1
	VAT1	0 1
	SNARE	0 1
✓	norepinephrine_sim...	0 1
	amine br translocat	0 1

[illegible]

EXPORT CSV

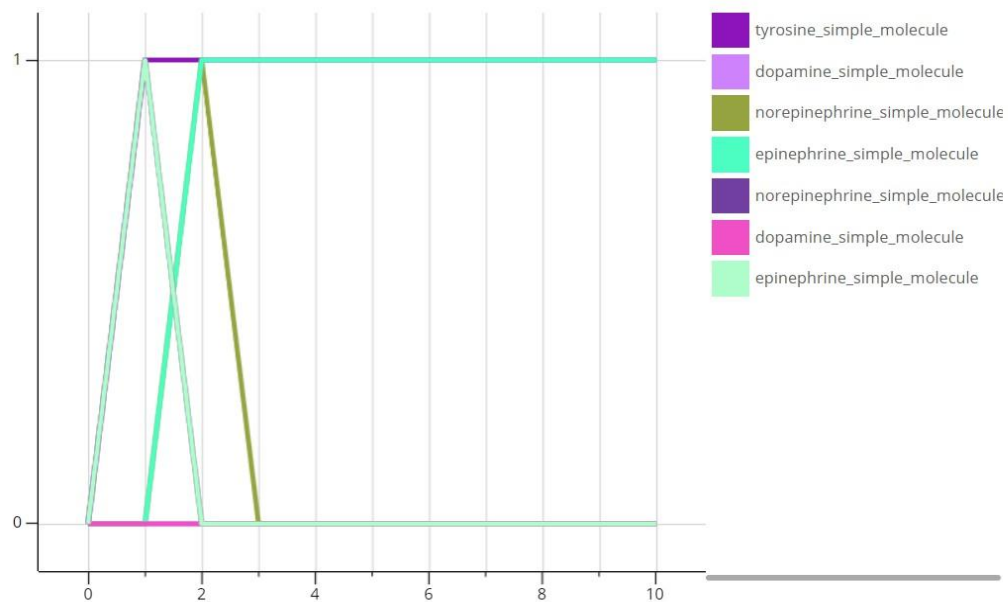
- 10

STEPS: 10

+ 10

Run

Simulation Graph



Bonus exercise for the CoLoMoTo notebook

Tape in your terminal the following command: colomoto-docker

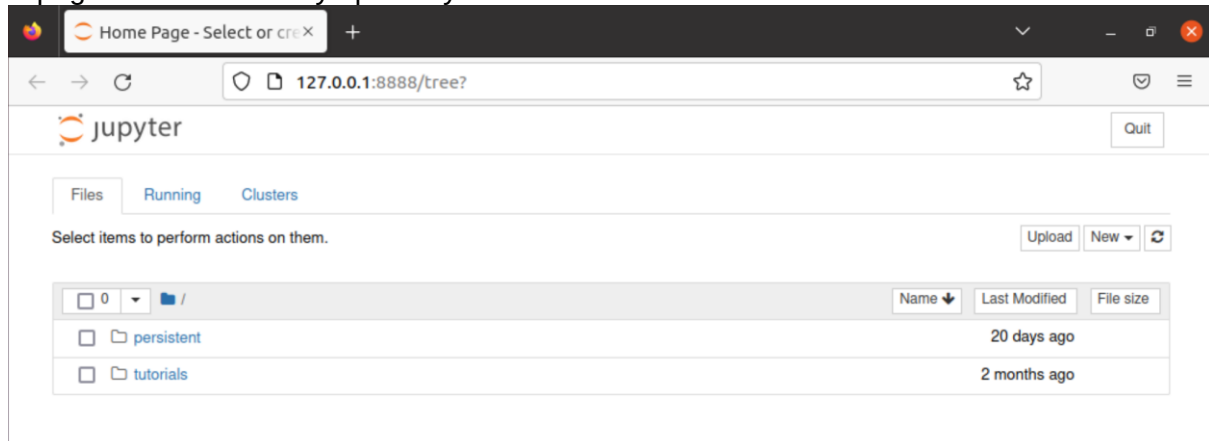
```

manager@CompSysBio22: ~
colomoto-docker
Client: Docker Engine - Community
 Version:      20.10.21
 API version:  1.41
 Go version:   go1.18.7
 Git commit:   baeda1f
 Built:        Tue Oct 25 18:02:21 2022
 OS/Arch:      linux/amd64
 Context:      default
 Experimental: true

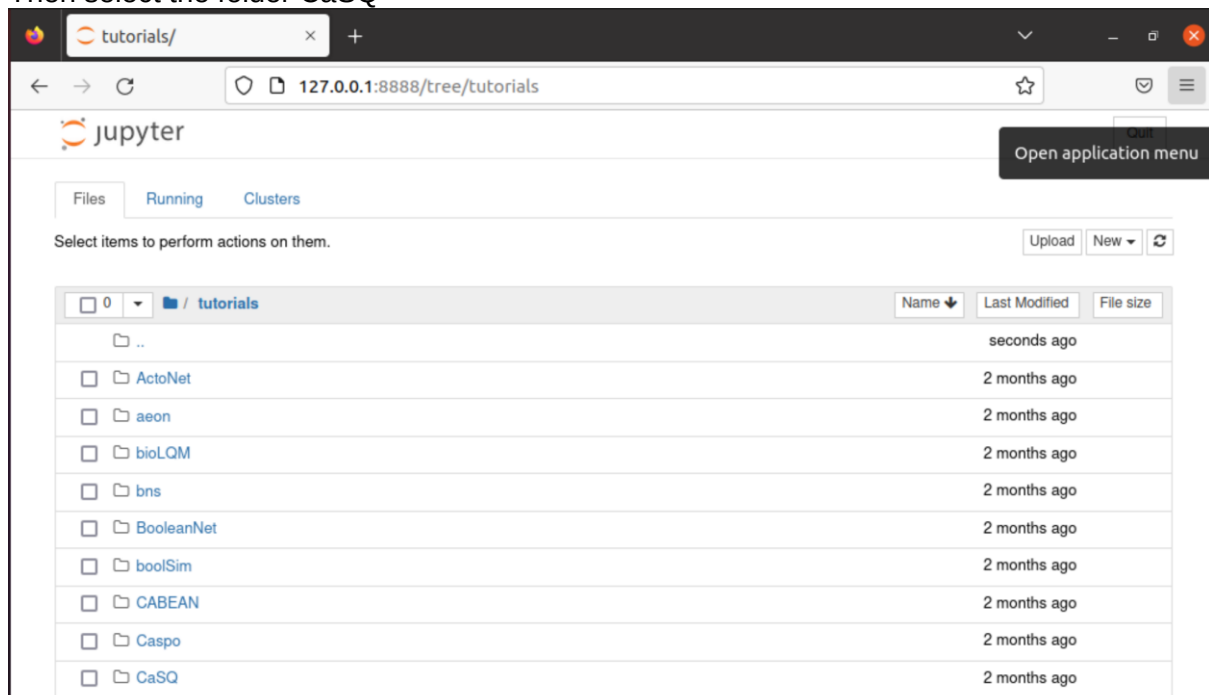
Server: Docker Engine - Community
 Engine:
  Version:      20.10.21
  API version:  1.41 (minimum version 1.12)
  Go version:   go1.18.7
  Git commit:   3056208
  Built:        Tue Oct 25 18:00:04 2022

```

A page will automatically open in your browser. Click on the folder tutorials



Then select the folder CaSQ



Then select the jupyter notebook

Jupyter

Files Running Clusters

Select items to perform actions on them.

Upload New

	Name	Last Modified	File size
0	tutorials / CaSQ		
	..	il y a quelques secondes	
	CaSQ_from_CellDesigner_to_GINsim.ipynb	Actif il y a 7 mois	14.4 kB
	Apoptosis_VS_SSA_AN.xml	il y a 7 mois	129 kB
	map_mastocell.xml	il y a 7 mois	1.01 MB

You will now perform your analysis in a reproducible and automated way: To execute the commands you need to click on the black arrowheads as shown below:

Jupyter CaSQ_from_CellDesigner_to_GINsim (auto-sauvegardé)

File Edit View Insert Cell Kernel Widgets Help

Non fiable Python 3

Exécuter

Create an executable model with CaSQ

Entrée [1]:

```
import casq.celldesigner2qual as casq
from colomoto_jupyter import tabulate

# debug messages are enabled by default
casq.logger.disable("casq")
```

Entrée [2]:

```
# Load and simplify a cell designer map
info, width, height = casq.read_celldesigner("Apoptosis_VS_SSA_AN.xml")
casq.simplify_model(info)

# Write the SBML file
casq.write_qual("Apoptosis_VS_SSA_AN.sbml", info, width, height)
```

Execute the commands to obtain an sbml file

In the next two steps, you are going to import the sbml qual file to GINsim, another software for logical modelling and simulations and you will visualize your model

Load and view the model in GINsim

Entrée [3]:

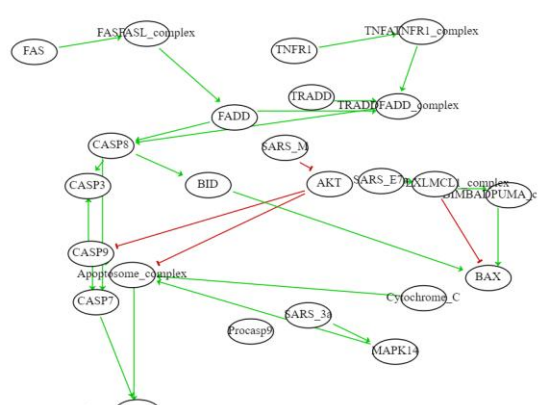
```
import biolqm
import ginsim
```

Entrée [4]:

```
m = biolqm.load("Apoptosis_VS_SSA_AN.sbml")
m = biolqm.sanitize(m)

lrg = biolqm.to_ginsim(m)
ginsim.show(lrg)
```

Out[4]:



Model imported in GINsim, the regulatory graph is depicted.

You can now click inside the space of the fourth command and then click on Insert and choose Insert cell below

Jupyter CaSQ_from_CellDesigner_to_GINsim (auto-sauvegardé) Python 3

File Edit View Insert Cell Kernel Widgets Help

Insert Cell Above
Insert Cell Below

ca sis_VS_SSA_AN.sbml", info, width, height)

Load and view the model in GINsim

```
Entrée [3]: import biolqm
import ginsim

Entrée [4]: m = biolqm.load("Apoptosis_VS_SSA_AN.sbml")
m = biolqm.sanitize(m)

lrg = biolqm.to_ginsim(m)
ginsim.show(lrg)
```

You will click twice to insert two cells at the end of the notebook



```
Entrée [ ]: 
```

```
Entrée [ ]: 
```

Inserting new cells to the notebook

In the first cell, you will type the following command and execute to obtain the stable states of the system.

```
Entrée [5]: fps = biolqm.fixpoints(m)
tabulate(fps)
```

Out[5]:

	FASFASL_complex	BIMBADPUMA_complex	Apoptosome_complex	TNFBTNFR1_complex	BCL2BCLXLML1_complex	TRADDADD_complex	TNFR1	F
0	0	0	0	0	0	0	0	0
1	0	0	0	0	0	0	0	0
2	0	0	0	0	0	0	0	0
3	0	0	0	0	0	0	0	0
4	0	0	0	0	0	0	0	0
...	...	...	...	...	...	...	...	...
251	1	1	1	1	1	1	1	1
252	1	1	1	1	1	1	1	1
253	1	1	1	1	1	1	1	1
254	1	1	1	1	1	1	1	1
255	1	1	1	1	1	1	1	1

256 rows x 24 columns

Calculating stable states (fixed points).

and in the second cell you will type:

```
Entrée [6]: import pandas as pd
pd.DataFrame(fps).to_csv("fixed_points.csv", index=False)
```

Saving the attractors in a csv file for further processing.

Key notions:

1. XML files of molecular maps encode Process Description diagrams (static)
2. CaSQ compresses the PD diagrams to AF (activity flow) and gives also logical rules that govern the regulation of the components.
3. SIF files contain only the structure of the Boolean models (AF, static) and can be used for topological analysis.
4. SBML qual files contain the executable model (structure plus logical rules) and can be used for *in silico* simulations (dynamic analysis).